EXPERIMENT No 11:
Building an Automatic Irrigation System

Student Guide

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Objective:

The purpose of this experiment is to build an automated irrigation system using an Arduino microcontroller. The system will detect soil moisture levels and water the plants when the soil is dry

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Materials Needed:

- **Arduino Uno** (or any compatible microcontroller)
- **Soil Moisture Sensor**
- **5V Relay Module**
- **Water Pump**
- **Power Supply (for the water pump)**
- **Breadboard and Jumper Wires**
- **Water Reservoir** (container for water)
- **Connecting pipes or tubing for water delivery**

- **Optional**: LCD Display, Temperature and Humidity Sensor (for more advanced functionality)

Project Overview:

In this project, you will create an automated system that:

- Monitors the moisture level in the soil.
- Automatically waters the plants when the soil moisture is below a set threshold.
- Turns off the water pump when the soil is sufficiently watered.

Steps to Build the System:

1. Connecting the Components:

- **Soil Moisture Sensor**: Connect the analog output of the soil moisture sensor to one of the analog pins (e.g., A0) on the Arduino.

- **Relay Module**: Connect the IN pin of the relay to a digital pin (e.g., D7). The relay controls the water pump.
- **Water Pump**: Connect the pump to a power source via the relay. The relay acts as a switch to turn the pump on and off.
- **Power Supply**: The Arduino can be powered via USB, but the water pump requires an external power supply (based on its specifications).

2. Circuit Diagram:

The following diagram shows the basic connections:

- **Soil Moisture Sensor** → Arduino (A0 pin)
- **Relay Module** → Arduino (D7 pin)
- **Water Pump** → Power Supply via Relay

3. Programming the Arduino:

- Open the **Arduino IDE** or **Mixly** (block-based programming).

- Write a program to monitor the soil moisture and control the pump based on sensor readings.

Basic Arduino Code:

sketch_oct10a | Arduino IDE 2.3.2

File Edit Sketch Tools Help

Arduino Uno

sketch_oct10a.ino

```
int moistureSensorPin = A0;  // Analog pin connected to soil moisture sensor
int relayPin = 7;            // Digital pin connected to the relay
int sensorValue = 0;         // Variable to store sensor reading
int threshold = 400;         // Moisture threshold (adjust based on testing)

void setup() {
  pinMode(relayPin, OUTPUT);     // Set relay pin as output
  digitalWrite(relayPin, HIGH);  // Initially turn off the pump
  Serial.begin(9600);            // Start serial communication
}
void loop() {
  sensorValue = analogRead(moistureSensorPin);  // Read soil moisture level
  Serial.print("Moisture Level: ");
  Serial.println(sensorValue);
  if (sensorValue < threshold) {
    digitalWrite(relayPin, LOW);   // Turn on the pump
    Serial.println("Pump ON");
  } else {
    digitalWrite(relayPin, HIGH);  // Turn off the pump
    Serial.println("Pump OFF");
  }
  delay(2000);  // Wait for 2 seconds before checking again
}
```

4. Testing the System:

- Insert the soil moisture sensor into the soil.
- When the soil is dry (sensor reading is low), the water pump will activate and water the plants.
- When the soil becomes wet (sensor reading is high), the water pump will turn off.

5. Optional Enhancements:

- **LCD Display**: Add an LCD display to show the current moisture level.
- **Temperature and Humidity Sensor**: Add environmental monitoring features to make the system smarter.

Understanding the Code:

- **moistureSensorPin**: The pin where the soil moisture sensor is connected.

- **relayPin**: The pin that controls the relay, which switches the pump on or off.
- **sensorValue**: Reads the moisture level from the sensor.
- **threshold**: The moisture level below which the pump will turn on.
- The loop() function continuously checks the sensor and activates the pump when necessary.

Project Outcome:

By the end of this project, you will have built an automated irrigation system that helps in efficient water use. This system can be expanded with more advanced features for monitoring environmental conditions or integrating a scheduling system.

Key Questions for Reflection:

1. What would happen if the soil moisture sensor is not calibrated correctly?
2. How can we expand the system to control multiple plants or garden sections?
3. What are some real-world applications of this type of automation in agriculture?

This project introduces automation, sensor interfacing, and practical programming. It's a real-world application that can be scaled up for large farms or small home gardens. Keep experimenting to see how you can improve it!